

TASK MANAGER APPLICATION

Task manager is a simple console-based application written in Java, which allows users to manage personal tasks. It contains the following features: creating, editing and selecting tasks, searching tasks and sorting tasks, saving tasks to files (in TXT or JSON format), and loading tasks from files.

MAIN MENU

When the Task Manager application is started, a simple main menu appears in the console. This menu displays the available options of the application and the corresponding commands that users need to enter to use them.

All available features are explained in the following sections. For better understanding, see the image below showing the main menu of the application.

```
----- Task Manager MAIN MENU -----  
1) To ADD new task, enter 'a' and press Enter  
2) To DELETE existing task, enter 'd' and press Enter  
3) To EDIT existing tasks, enter 'e' and press Enter  
4) To SAVE existing tasks as text file, enter 'stxt' and press Enter  
5) To LOAD a list of tasks from text file, enter 'ltxt' and press Enter  
6) To SAVE existing tasks as JSON file, enter 'sjson' and press Enter  
7) To LOAD a list of tasks from JSON file, enter 'ljson' and press Enter  
8) To PRINT all tasks, enter 'p' and press Enter  
9) To SORT and PRINT tasks according to PRIORITY, enter 'pp' and press Enter  
10) To SORT and PRINT tasks according to DEADLINE, enter 'pd' and press Enter  
11) To PRINT tasks with defined TYPE, enter 'pt' and press Enter  
12) To SEARCH tasks with defined KEYWORD, enter 'pk' and press Enter  
13) To PRINT tasks STATISTICS, enter 'stat' and press Enter  
14) To END this program, enter 'q' and press Enter
```

TASK MANAGEMENT

Add task

A new task can be created by selecting the **ADD** option. To start the process of adding a new task, the user needs to enter the key 'a' into the console and press the 'Enter' key. The process of adding a task contains the following steps:

- 1) **Name:** The application prompts the user to enter the name of the new task. The user must specify the name and press the 'Enter' key. If the entered name already exists (another task has the same name), the application notifies the user and offers the option to either enter a different name or generate an automatic name. The task name is the only mandatory parameter.
- 2) **Description:** The application allows the user to add a description to the task. If the user does not wish to add a description, this step can be skipped by pressing the 'Enter' key without entering any text.
- 3) **Deadline:** Next, the application offers the option to add a deadline to the task. The deadline can also be skipped by pressing the 'Enter' key. Deadlines must be entered in the correct format: DD.MM.YYYY. If the entered deadline has an incorrect format, the application notifies the user and offers the option to try again or continue without specifying a deadline.
- 4) **Priority:** Next, the application offers the option to assign a priority to the task. Priority is represented by an integer value, where: 0 means no priority, 1 represents the highest priority, and higher numbers represent lower priorities. This step can also be skipped by pressing the 'Enter' key.
- 5) **Type:** Finally, the application allows the user to assign a type (category) to the new task. Types are useful for grouping tasks, for example to school homework, work duties, or personal tasks such as chores or groceries. This step can be also skipped by pressing the 'Enter' key without specifying a type.
- 6) **Completion state:** The last parameter of every task is the completion state. Newly created tasks are automatically marked as incomplete. This state can later be changed when the task is completed.

The whole process of adding a new task is shown in the image below.

```
a
Write name of new task and press Enter:
Task Manager Application
If you would like to add a description of this task, write it down and press Enter, otherwise just press Enter:
Write user documentation to Task Manager Application
If you would like to add a deadline to this task, enter deadline in format 'DD.MM.YYYY' and press Enter, otherwise just press Enter:
22.05.2026
If you would like to add this task a priority, enter a number (lower number = higher priority) and press Enter, otherwise just press Enter (no priority defaults to 0):
1
If you would like to add this task's type (work/personal/school subject), enter a type and press Enter, otherwise, just press Enter:
school
Setting task parameters...
Task successfully created!

Created task:

Name: Task Manager Application
Description: Write user documentation to Task Manager Application
Deadline: 22.05.2026 (in 7 days)
Priority: 1
Type: school
Completed: incomplete
```

Delete task

An existing task can be deleted by selecting the **DELETE** option. To start the process, the user needs to enter the key 'd' and press the 'Enter' key. The application then prompts the user to enter the name of the task to be deleted. If the entered name does not exist, the application notifies the user and returns to the main menu. Otherwise, the application displays a message confirming successful deletion. These two scenarios are shown in the image below. For this example, assume that the manager contains a task named "Task Manager Application", but no task named "Task Manager".

```
d
Enter name of task you would like to delete and press Enter:
Task Manager
No task with such name was found!

----- Task Manager MAIN MENU -----
1) To ADD new task, enter 'a' and press Enter
2) To DELETE existing task, enter 'd' and press Enter
3) To EDIT existing tasks, enter 'e' and press Enter
4) To SAVE existing tasks as text file, enter 'stxt' and press Enter
5) To LOAD a list of tasks from text file, enter 'ltxt' and press Enter
6) To SAVE existing tasks as JSON file, enter 'sjson' and press Enter
7) To LOAD a list of tasks from JSON file, enter 'ljson' and press Enter
8) To PRINT all tasks, enter 'p' and press Enter
9) To SORT and PRINT tasks according to PRIORITY, enter 'pp' and press Enter
10) To SORT and PRINT tasks according to DEADLINE, enter 'pd' and press Enter
11) To PRINT tasks with defined TYPE, enter 'pt' and press Enter
12) To SEARCH tasks with defined KEYWORD, enter 'pk' and press Enter
13) To PRINT tasks STATISTICS, enter 'stat' and press Enter
14) To END this program, enter 'q' and press Enter

d
Enter name of task you would like to delete and press Enter:
Task Manager Application
Task Task Manager Application successfully deleted!
```

Edit task

An existing task can be edited by selecting the **EDIT** option. To start the editing process, the user needs to enter the key 'e' and press the 'Enter' key. The application then prompts the user to enter the name of the task to edit. If the entered name does not exist, the application notifies the user and returns to the main menu. If a valid task name is entered, the application opens the edit

menu, where the user can choose which parameter to edit. The process of selecting a task and opening the edit menu is shown in the image below.

```
e
Enter name of task you would like to edit and press Enter:
Task Manager
No task with such name was found!

----- Task Manager MAIN MENU -----
1) To ADD new task, enter 'a' and press Enter
2) To DELETE existing task, enter 'd' and press Enter
3) To EDIT existing tasks, enter 'e' and press Enter
4) To SAVE existing tasks as text file, enter 'stxt' and press Enter
5) To LOAD a list of tasks from text file, enter 'ltxt' and press Enter
6) To SAVE existing tasks as JSON file, enter 'sjson' and press Enter
7) To LOAD a list of tasks from JSON file, enter 'ljson' and press Enter
8) To PRINT all tasks, enter 'p' and press Enter
9) To SORT and PRINT tasks according to PRIORITY, enter 'pp' and press Enter
10) To SORT and PRINT tasks according to DEADLINE, enter 'pd' and press Enter
11) To PRINT tasks with defined TYPE, enter 'pt' and press Enter
12) To SEARCH tasks with defined KEYWORD, enter 'pk' and press Enter
13) To PRINT tasks STATISTICS, enter 'stat' and press Enter
14) To END this program, enter 'q' and press Enter

e
Enter name of task you would like to edit and press Enter:
Task Manager Application
----- Task Manager EDIT MENU -----
1) To edit task's NAME, enter 'n' and press Enter
2) To edit task's DESCRIPTION, enter 'ds' and press Enter
3) To edit task's DEADLINE, enter 'dl' and press Enter
4) To edit task's PRIORITY, enter 'p' and press Enter
5) To edit task's TYPE, enter 't' and press Enter
6) To edit task's COMPLETENESS, enter 'c' and press Enter
7) To EXIT edit mode, enter 'q' and press Enter
```

- 1) **Edit name:** The task name can be edited by entering the key 'n' and pressing the 'Enter' key in the edit menu. The application displays the current name and prompts the user to enter a new one. As with task creation, duplicate names are not allowed. If the user enters an existing name, the application notifies the user and offers the option to either enter a different name or keep the current one using the 'q' command.

- 2) **Edit description:** The task description can be edited by entering the key 'ds' and pressing the 'Enter' key. The application displays the current description and allows the user to:
 - enter a new description
 - delete the current description using the 'd' command
 - keep the current description using the 'q' command
- 3) **Edit deadline:** The task deadline can be edited by entering the key 'dl' and pressing the 'Enter' key. The application displays the current deadline and allows the user to:
 - enter a new deadline in the correct format
 - delete the current deadline using the 'd' command
 - keep the current deadline using the 'q' command
- 4) **Edit priority:** The task priority can be edited by entering the key 'p' and pressing the 'Enter' key. The application displays the current priority and allows the user to:
 - enter a new priority
 - delete the current priority using the 'd' command
 - keep the current priority using the 'q' command
- 5) **Edit type:** The task type can be edited by entering the key 't' and pressing the 'Enter' key. The application displays the current type and allows the user to:
 - enter a new type
 - delete the current type using the 'd' command
 - keep the current type using the 'q' command
- 6) **Edit completion state:** The task completion state can be edited by entering the key 'c' and pressing the 'Enter' key. If the task is incomplete, the application offers the option to mark it as completed by pressing 'Enter' or leave it unchanged using the 'q' command. The same logic applies in the opposite direction. If the user changes the task state from incomplete to completed, the application additionally offers the option to delete the completed task. The user can choose:
 - 'yes' – delete the task. If the user chooses to delete the task, the application returns to the main menu.
 - 'no' – keep the task
- 7) **Exit edit mode:** The edit mode can be exited using the 'q' command.

FILE OPERATIONS

Save tasks to TXT file

Tasks created in the application can be saved to a TXT file. To start the saving process, the user needs to enter the command 'stxt' and press the 'Enter' key. The application then prompts the user to enter a file name or generate an automatic file name by pressing the 'Enter' key. If the

provided file name is valid, or if an automatic name is generated, the application saves the tasks and displays a confirmation message including the file path. The application also checks for duplicate file names to prevent overwriting existing files. If a duplicate name is detected, the application displays an error message and prompts the user to enter a different name.

Save tasks to JSON file

Tasks can also be saved to a JSON file by entering the command 'sjson' and pressing the 'Enter' key. The saving process is identical to saving tasks to a TXT file.

For both saving mechanisms, if the saving process fails, the application displays an error message and returns to the main menu. The image below shows successful saving with an automatically generated file name and a situation where a duplicate file name is detected.

```
stxt
Please enter name under which this file will be saved or press Enter to select automatic name:

Generating automatic name...
Tasks successfully saved to: C:\Users\Terezka\School\UK\Java\TaskManager\data\tasks1.txt
----- Task Manager MAIN MENU -----
1) To ADD new task, enter 'a' and press Enter
2) To DELETE existing task, enter 'd' and press Enter
3) To EDIT existing tasks, enter 'e' and press Enter
4) To SAVE existing tasks as text file, enter 'stxt' and press Enter
5) To LOAD a list of tasks from text file, enter 'ltxt' and press Enter
6) To SAVE existing tasks as JSON file, enter 'sjson' and press Enter
7) To LOAD a list of tasks from JSON file, enter 'ljson' and press Enter
8) To PRINT all tasks, enter 'p' and press Enter
9) To SORT and PRINT tasks according to PRIORITY, enter 'pp' and press Enter
10) To SORT and PRINT tasks according to DEADLINE, enter 'pd' and press Enter
11) To PRINT tasks with defined TYPE, enter 'pt' and press Enter
12) To SEARCH tasks with defined KEYWORD, enter 'pk' and press Enter
13) To PRINT tasks STATISTICS, enter 'stat' and press Enter
14) To END this program, enter 'q' and press Enter

stxt
Please enter name under which this file will be saved or press Enter to select automatic name:
tasks1
This filename already exists. Please enter different name or press Enter to select automatic name:

Generating automatic name...
Tasks successfully saved to: C:\Users\Terezka\School\UK\Java\TaskManager\data\tasks2.txt
```

Load tasks from TXT file

Tasks can be loaded from a TXT file by selecting the **LOAD** option. To start the loading process, the user needs to enter the command 'ltxt' and press the 'Enter' key. The application displays all available TXT files in the data directory and prompts the user to enter the name of

the file to load. If the entered file name is valid, the tasks are loaded and the application displays a confirmation message. If the entered file name does not exist, the application displays an error message and prompts the user to try again.

Load tasks from JSON file

Loading tasks from a JSON file works similarly to loading tasks from TXT files. The loading process is started using the command 'ljson'. For both loading mechanisms, if loading fails, the application displays an error message and returns to the main menu.

The image below demonstrates both unsuccessful and successful loading attempts.

```
ltxt
Enter name of file you would like to load. Available text files in directory 'data':
tasks1.txt tasks2.txt
tasks
File does not exist, please enter valid file name:
tasks1
Tasks successfully loaded!
----- Task Manager MAIN MENU -----
1) To ADD new task, enter 'a' and press Enter
2) To DELETE existing task, enter 'd' and press Enter
3) To EDIT existing tasks, enter 'e' and press Enter
4) To SAVE existing tasks as text file, enter 'stxt' and press Enter
5) To LOAD a list of tasks from text file, enter 'ltxt' and press Enter
6) To SAVE existing tasks as JSON file, enter 'sjson' and press Enter
7) To LOAD a list of tasks from JSON file, enter 'ljson' and press Enter
8) To PRINT all tasks, enter 'p' and press Enter
9) To SORT and PRINT tasks according to PRIORITY, enter 'pp' and press Enter
10) To SORT and PRINT tasks according to DEADLINE, enter 'pd' and press Enter
11) To PRINT tasks with defined TYPE, enter 'pt' and press Enter
12) To SEARCH tasks with defined KEYWORD, enter 'pk' and press Enter
13) To PRINT tasks STATISTICS, enter 'stat' and press Enter
14) To END this program, enter 'q' and press Enter
```

TASK VIEWING

Print all tasks

The application can display all existing tasks together with all their parameters. To use this feature, the user needs to enter the command 'p' and press the 'Enter' key. If no tasks exist, the application displays an appropriate message and returns to the main menu.

Sort tasks according to priority

The application can sort and display tasks according to priority. Tasks with higher priority appear first. Since lower numerical values represent higher priorities, tasks with lower priority numbers are displayed before tasks with higher numbers. To use this feature, the user needs to enter the command 'pp'. If no tasks exist, the application displays an appropriate message and returns to the main menu.

Sort tasks according to deadline

The application can also sort tasks according to their deadlines. To use this feature, the user needs to enter the command 'pd'. If no tasks exist, the application displays an appropriate message and returns to the main menu.

Print tasks with specified type

The application can display only tasks of a selected type. To use this feature, the user needs to enter the command 'pt'. The application then prompts the user to enter the desired type. If no tasks with the specified type exist, the application displays a message and offers the option to either try another type or return to the main menu using the 'q' command. Otherwise, all matching tasks are displayed.

Print tasks with defined keyword

The application also supports searching tasks using keywords. This feature can be used by entering the command 'pk'. The application then prompts the user to enter a keyword. If the keyword is not found in any task, the application displays a message and offers the option to either enter another keyword or return to the main menu using the 'q' command. If matching tasks are found, the application displays all tasks containing the keyword, with the keyword highlighted. For this example, assume that only one task exists — the task created in the earlier “Add task” section. The image below demonstrates both unsuccessful and successful keyword searches.

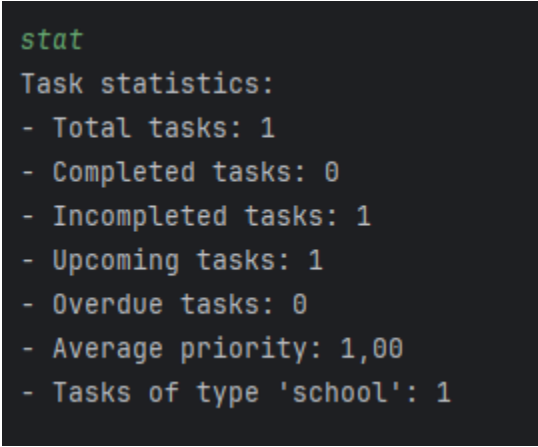
```
pk
Enter keyword you want to search for tasks by and press Enter:
Homework
Defined keyword does not exist. Please enter valid keyword, or enter 'q' and press Enter to go back to main menu:
Manager
----- TASKS CONTAINING KEYWORD 'Manager' -----
Name: Task Manager Application
Description: Write user documentation to Task Manager Application
Deadline: 22.05.2026 (in 7 days)
Priority: 1
Type: school
Completed: incompleted
-----
```


Print tasks statistics

The application also provides task statistics, including:

- total number of tasks
- completed tasks
- incomplete tasks
- overdue tasks
- upcoming tasks
- average priority
- task type distribution

To display task statistics, the user needs to enter the command 'stat'. If no tasks exist, the application displays an appropriate message and returns to the main menu. The image below shows statistics for the task created in the earlier "Add task" section.



```
stat
Task statistics:
- Total tasks: 1
- Completed tasks: 0
- Incompleted tasks: 1
- Upcoming tasks: 1
- Overdue tasks: 0
- Average priority: 1,00
- Tasks of type 'school': 1
```

EXIT APPLICATION

To exit the Task Manager Application, the user needs to enter the command 'q' and press the 'Enter' key. The application will then terminate.